List of Covered Topics (Midterm Scope)

Students are expected to understand the following concepts conceptually and mathematically:

- **Floating Point Arithmetic:** fp16 vs bf16 (mantissa vs exponent range), machine epsilon, efficient matrix multiplication algorithms.
- **Matrices:** normal, SPD, Gram, defective, invertible, identity, symmetric, diagonal, triangular.
- **Vector and Matrix Norms:** Equivalence of p-norms, constants depending on dimension n, unitary invariance of Frobenius ($\|\cdot\|_F$) and Spectral ($\|\cdot\|_2$) norms, counterexamples for $\|\cdot\|_1$ and $\|\cdot\|_\infty$.
- **Conditioning and Stability:** Definition of $\kappa(A)$, why $\kappa(A) \geq 1$, small determinant $\not\Rightarrow$ ill-conditioned, Gershgorin circle theorem basics.
- **Matrix Factorizations:** SVD, LU decomposition, Cholesky, QR decomposition, QR-algorithm, Cost of solving multiple right-hand sides.
- **Least Squares:** Normal equations (and why they square the condition number), QR-based solvers, SVD pseudoinverse.
- **SVD and Low-Rank Approximation:** Eckart-Young-Mirsky theorem (for both $\|\cdot\|_2$ and $\|\cdot\|_F$), truncated SVD.
- **Eigenvalues:** Spectral properties of triangular/Hermitian matrices, Power method to find leading (maximum absolute value) eigenvalue and eigenvector, Power method requirements (strictly dominant eigenvalue), dangers of characteristic polynomials.

Examples of questions (with expected solutions)

Note that the number of questions in real mid term is more then presented here. The concepts are taken from lectures and HWs. Expected to be one proof task from the HW.

1. (1 point) Which of the following statements about matrix conditioning is **always true**?
   A. A very small determinant $\det(A)$ implies that A is close to singular.
   B. The condition number satisfies $\kappa_2(A) \geq 1$.
   C. The condition number is always given by $\kappa_2(A) = \max_i |\lambda_i| / \min_i |\lambda_i|$.
   D. If A is invertible, it always admits an LU decomposition without row permutations.

   Solution: B
   As $\kappa_2(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$, the statement follows. A small determinant can simply be due to scaling (e.g., $0.1I$ has det $= 10^{-n}$ but $\kappa = 1$). The eigenvalue ratio formula only holds for normal matrices. Permutations are often needed (e.g., $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$).

2. (1 point) When solving a least-squares problem using the Normal Equations, what happens to numerical stability?

   Solution: It squares the condition number, $\kappa_2(A^\top A) = (\kappa_2(A))^2$, potentially losing half the precision.

The singular values of $A^\top A$ are the squares of the singular values of A.

Thus, $\frac{\sigma_{\max}(A^\top A)}{\sigma_{\min}(A^\top A)} = \frac{(\sigma_{\max}(A))^2}{(\sigma_{\min}(A))^2} = \kappa_2(A)^2$.

This amplifies rounding errors significantly for ill-conditioned A.

3. (1 point) Let $A \in \mathbb{R}^{m\times n}$ have singular values $\sigma_1 \geq \cdots \geq \sigma_p \geq 0$, $p = \min(m,n)$, and let $r < p$. According to the Eckart-Young-Mirsky theorem, what is $\min_{\operatorname{rank} B \leq r} \|A - B\|_F$, and which matrix attains it?

Solution: $\min\limits_{\operatorname{rank} B \leq r} \|A - B\|_F = \sqrt{\sum\limits_{i=r+1}^{p} \sigma_i^2}$, attained by the truncated SVD $A_r = \sum_{i=1}^{r} \sigma_i u_i v_i^\top$.

$A - A_r = \sum_{i=r+1}^{p} \sigma_i u_i v_i^\top$ keeps only the discarded terms, and by unitary invariance its Frobenius norm is the square root of the sum of the squared discarded singular values. The theorem states that no matrix of rank at most r does better.

4. (1 point) The Frobenius and spectral norms are unitarily invariant. Give an orthogonal matrix Q and a matrix A with $\|QA\|_1 \neq \|A\|_1$, where $\|\cdot\|_1$ is the matrix norm induced by the vector 1-norm.

Solution: I think... $A = I_2$ and $Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ this is rotation matrix by 45°.

The induced 1-norm is the maximal absolute column sum. $\|I_2\|_1 = 1$, while every column of Q has absolute sum $2/\sqrt{2} = \sqrt{2}$, so $\|QA\|_1 = \sqrt{2}$. Geometrically rotation preserves the Euclidean length, but it turns e_1 (1-norm 1) into $(1,1)^\top/\sqrt{2}$ (1-norm $\sqrt{2}$). The same example works for $\|\cdot\|_\infty$.

5. (3 points) **Prove that $\|UAV\|_F = \|A\|_F$ for unitary U and V.**

Solution: By definition, $\|X\|_F^2 = \operatorname{trace}(X^*X)$. Let $X = UAV$. Then:

$$\|UAV\|_F^2 = \operatorname{trace}((UAV)^*(UAV)) = \operatorname{trace}(V^*A^*U^*UAV)$$

Since U is unitary, $U^*U = I$. This simplifies to:

$$\operatorname{trace}(V^*A^*AV)$$

Using the cyclic property of the trace, $\operatorname{trace}(XYZ) = \operatorname{trace}(ZXY)$, we can move V to the front:

$$\operatorname{trace}(VV^*A^*A)$$

Since V is unitary, $VV^* = I$. Thus:

$$\operatorname{trace}(A^*A) = \|A\|_F^2$$

Taking the square root of both sides yields $\|UAV\|_F = \|A\|_F$. □ (3 pts: 1 pt for def, 1 pt for $U^*U = I$, 1 pt for cyclic trace and $VV^* = I$)